

Assignment 2

Lab 4: Advanced Missing Data Handling

UCI Adult / Census Income Dataset

This dataset is particularly suitable because it contains a mixture of numeric and categorical variables such as age, work class, education, occupation, working hours, income, etc.

```
> library(naniar)
> library(skimmr)

Attaching package: 'skimmr'

The following object is masked from 'package:naniar':
  n_complete

> col_names <- c(
+   "age", "workclass", "fnlwgt", "education", "education_num",
+   "marital_status", "occupation", "relationship", "race",
+   "sex", "capital_gain", "capital_loss", "hours_per_week",
+   "native_country", "income"
+ )
> adult <- read.csv(
+   "adult.data",
+   header = FALSE,
+   col.names = col_names,
+   strip.white = TRUE,
+   na.strings = "?"
+ )
> str(adult)
'data.frame':   32561 obs. of  15 variables:
 $ age          : int   39 50 38 53 28 37 49 52 31 42 ...
 $ workclass    : chr   "State-gov" "Self-emp-not-inc" "Private" "Private" ...
 $ fnlwgt       : int  77516 83311 215646 234721 338409 284582 160187 209642 45781 159449 ...
 $ education    : chr   "Bachelors" "Bachelors" "HS-grad" "11th" ...
 $ education_num : int   13 13 9 7 13 14 5 9 14 13 ...
 $ marital_status : chr   "Never-married" "Married-civ-spouse" "Divorced" "Married-civ-spouse" ...
 $ occupation   : chr   "Adm-clerical" "Exec-managerial" "Handlers-cleaners" "Handlers-cleaners" ...
 $ relationship : chr   "Not-in-family" "Husband" "Not-in-family" "Husband" ...
 $ race         : chr   "White" "White" "White" "Black" ...
 $ sex          : chr   "Male" "Male" "Male" "Male" ...
 $ capital_gain  : int   2174 0 0 0 0 0 0 0 14084 5178 ...
 $ capital_loss  : int    0 0 0 0 0 0 0 0 0 0 ...
 $ hours_per_week : int   40 13 40 40 40 40 16 45 50 40 ...
 $ native_country : chr   "United-States" "United-States" "United-States" "United-States" ...
 $ income       : chr   "<=50K" "<=50K" "<=50K" "<=50K" ...
> set.seed(123)
```

```
> adult$age[c(5,20)] <- NA
> adult$capital_gain[c(8,15)] <- NaN
> adult$workclass[c(12,18)] <- ""
> adult$age[c(30,45)] <- 999
> # Demonstrate NULL
> temp <- NULL
> # Save original dataset
> adult_before <- adult
> cat("===== BEFORE CLEANING =====\n\n")
```

```

===== BEFORE CLEANING =====
> cat("NA Values\n")
NA Values
> print(colSums(is.na(adult_before)))
      age      workclass      fnlwgt      education      education_num      marital_status      occupation
      2      1836      0      0      0      0      1843
relationship      race      sex      capital_gain      capital_loss      hours_per_week      native_country
      0      0      0      2      0      0      583
      income
      0
> cat("\nNaN Values\n")

NaN Values
> print(sapply(adult_before, function(x)
+ {
+   if(is.numeric(x))
+     sum(is.nan(x))
+   else
+     0
+ })))
      age      workclass      fnlwgt      education      education_num      marital_status      occupation
      0      0      0      0      0      0      0
relationship      race      sex      capital_gain      capital_loss      hours_per_week      native_country
      0      0      0      2      0      0      0
      income
      0
> cat("\nNULL Object\n")

NULL Object
> print(is.null(temp))
[1] TRUE
> cat("\nBlank Strings\n")

Blank Strings
> print(sapply(adult_before, function(x)
+ {
+   if(is.character(x))
+     sum(x == "", na.rm = TRUE)
+   else
+     0
+ })))
      age      workclass      fnlwgt      education      education_num      marital_status      occupation
      0      2      0      0      0      0      0
relationship      race      sex      capital_gain      capital_loss      hours_per_week      native_country
      0      0      0      0      0      0      0
      income
      0
> cat("\nImpossible Age Values\n")

Impossible Age Values
> print(sum(adult_before$age == 999, na.rm = TRUE))
[1] 2
> cat("\nVariable-wise Missing Summary\n")

Variable-wise Missing Summary
> print(miss_var_summary(adult_before))
# A tibble: 15 x 3
  variable      n_miss pct_miss
  <chr>      <int>    <num>
1 occupation    1843  5.66
2 workclass    1836  5.64
3 native_country    583  1.79
4 age           2  0.00614
5 capital_gain     2  0.00614
6 fnlwgt         0  0
7 education       0  0
8 education_num   0  0
9 marital_status  0  0
10 relationship   0  0
11 race          0  0
12 sex           0  0
13 capital_loss   0  0
14 hours_per_week 0  0
15 income        0  0

```

```

+   0
+ })))
      age      workclass      fnlwgt      education      education_num      marital_status      occupation
      0      2      0      0      0      0      0
relationship      race      sex      capital_gain      capital_loss      hours_per_week      native_country
      0      0      0      0      0      0      0
      income
      0
> cat("\nImpossible Age Values\n")

Impossible Age Values
> print(sum(adult_before$age == 999, na.rm = TRUE))
[1] 2
> cat("\nVariable-wise Missing Summary\n")

Variable-wise Missing Summary
> print(miss_var_summary(adult_before))
# A tibble: 15 x 3
  variable      n_miss pct_miss
  <chr>      <int>    <num>
1 occupation    1843  5.66
2 workclass    1836  5.64
3 native_country    583  1.79
4 age           2  0.00614
5 capital_gain     2  0.00614
6 fnlwgt         0  0
7 education       0  0
8 education_num   0  0
9 marital_status  0  0
10 relationship   0  0
11 race          0  0
12 sex           0  0
13 capital_loss   0  0
14 hours_per_week 0  0
15 income        0  0

```

```

> # Missingness visualization
> vis_miss(adult_before)
> adult <- adult_before
> # Convert impossible age values to NA
> adult$age[adult$age == 999] <- NA
> # Replace blank strings
> character_columns <- names(adult)[sapply(adult, is.character)]
> for(col in character_columns)
+ {
+   adult[[col]][adult[[col]] == ""] <- "Unknown"
+ }
> # Remove rows containing NaN
> numeric_columns <- names(adult)[sapply(adult, is.numeric)]
> for(col in numeric_columns)
+ {
+   adult <- adult[!is.nan(adult[[col]]), ]
+ }
> median_impute <- function(x)
+ {
+   median_value <- median(x, na.rm = TRUE)
+   x[is.na(x)] <- median_value
+   return(x)
+ }
> # Apply to all numeric columns
> numeric_columns <- names(adult)[sapply(adult, is.numeric)]
> for(col in numeric_columns)
+ {
+   adult[[col]] <- median_impute(adult[[col]])
+ }
> complete_rows <- complete.cases(adult)
> cat("\nComplete Rows :", sum(complete_rows), "\n")

```

Complete Rows : 30161

```

> cat("Incomplete Rows :", sum(!complete_rows), "\n")

```

Incomplete Rows : 2398

```

===== AFTER CLEANING =====
> cat("NA Values\n")
NA Values
> print(colSums(is.na(adult)))
      age      workclass      fnlwgt      education      education_num      marital_status      occupation
      0      1836      0      0      0      0      1843
relationship      race      sex      capital_gain      capital_loss      hours_per_week      native_country
      0      0      0      0      0      0      582
income
      0
> cat("\nMissing Summary\n")

Missing Summary
> print(miss_var_summary(adult))
# A tibble: 15 x 3
  variable      n_miss pct_miss
  <chr>      <int>    <num>
1 occupation    1843    5.66
2 workclass    1836    5.64
3 native_country    582    1.79
4 age           0      0
5 fnlwgt        0      0
6 education     0      0
7 education_num  0      0
8 marital_status 0      0
9 relationship   0      0
10 race          0      0
11 sex           0      0
12 capital_gain  0      0
13 capital_loss  0      0
14 hours_per_week 0      0
15 income       0      0
> vis_miss(adult)
> cat("\nValidation Results\n")

```

Validation Results

```

> cat("Impossible Age Values Remaining :",
+     sum(adult$age == 999), "\n")
Impossible Age Values Remaining : 0
> cat("Missing Age Values :",
+     sum(is.na(adult$age)), "\n")
Missing Age Values : 0
> cat("Blank Workclass Values :",
+     sum(adult$workclass == ""), "\n")
Blank Workclass Values : NA
> cat("Complete Cases :",
+     sum(complete.cases(adult)), "\n")
Complete Cases : 30161
> cat("Incomplete Cases :",
+     sum(!complete.cases(adult)), "\n")
Incomplete Cases : 2398
> # Comprehensive summary
> skim(adult)

```

```

— Data Summary —

```

Name	adult
Number of rows	32559
Number of columns	15
Column type frequency:	
character	9
numeric	6
Group variables	None

```
— Variable type: character —
skim_variable  n_missing complete_rate min max empty n_unique whitespace
1 workclass    1836      0.944  7  16    0      9      0
2 education    0        1      3  12    0     16      0
3 marital_status 0        1      7  21    0      7      0
4 occupation   1843    0.943  5  17    0     14      0
5 relationship 0        1      4  14    0      6      0
6 race         0        1      5  18    0      5      0
7 sex         0        1      4   6    0      2      0
8 native_country 582    0.982  4  26    0     41      0
9 income       0        1      4   5    0      2      0

— Variable type: numeric —
skim_variable  n_missing complete_rate  mean    sd    p0    p25    p50    p75    p100 hist
1 age          0          1    38.6   13.6   17    28    37    48    90
2 fnlwgt       0          1 189780. 105552. 12285 117822. 178356 237058 1484705
3 education_num 0          1    10.1    2.57    1     9    10    12    16
4 capital_gain 0          1   1078.   7386.    0     0     0     0   99999
5 capital_loss 0          1    87.3   403.    0     0     0     0   4356
6 hours_per_week 0          1   40.4   12.3    1    40    40    45    99
```

```
> write.csv(
+   adult,
+   "cleaned_adult_data.csv",
+   row.names = FALSE
+ )
> cat("\nCleaning Completed Successfully!\n")
```

Cleaning Completed Successfully!

